DeepLearning.AI

Math for Machine Learning

Linear algebra - Week 2

Solving systems of equations

Matrix row reduction

Row operations that preserve singularity

Row-reduced echelon form

Row echelon form

Rank of a matrix

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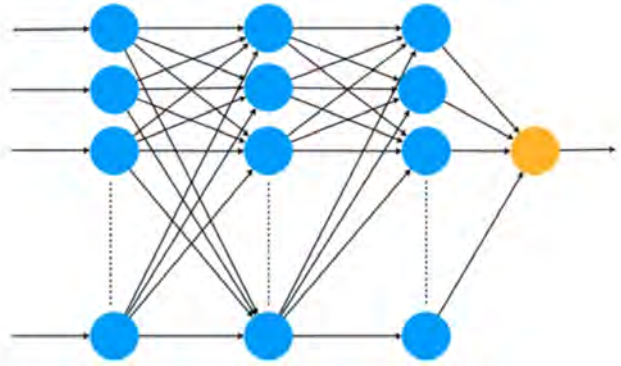
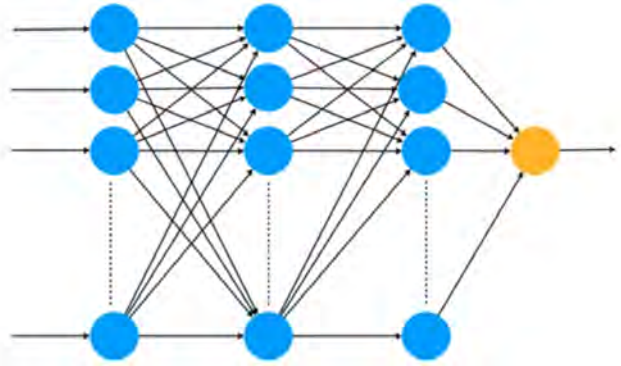
Solving System of Linear Equations

Machine learning motivation

Neural networks - Matrix operations

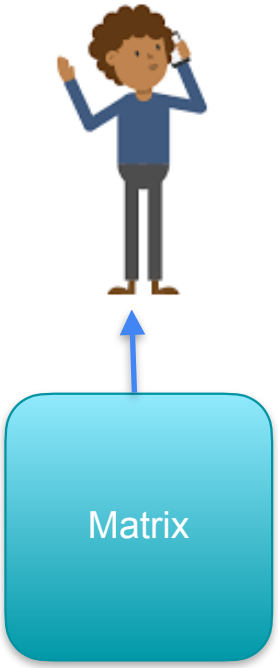


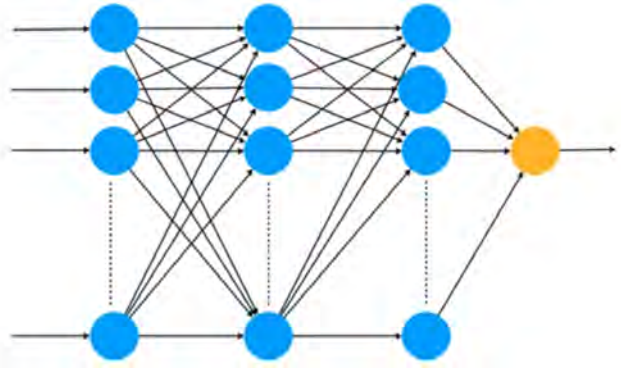
Neural networks - Matrix operations



"Hello! Welcome to Math for Machine Learning!"

Neural networks - Matrix operations





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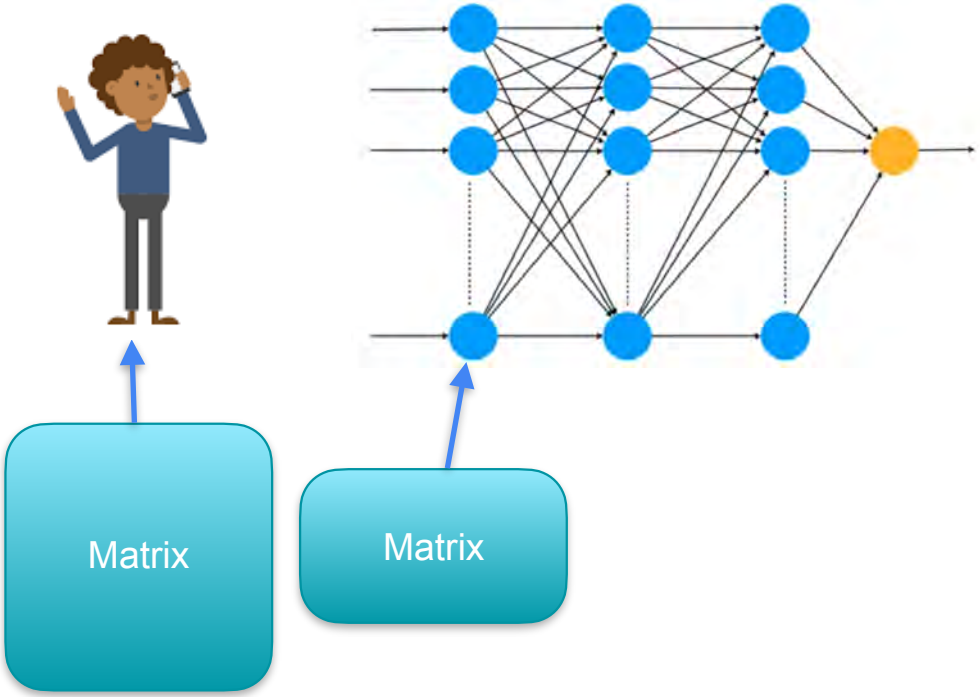
Neural networks - Matrix operations



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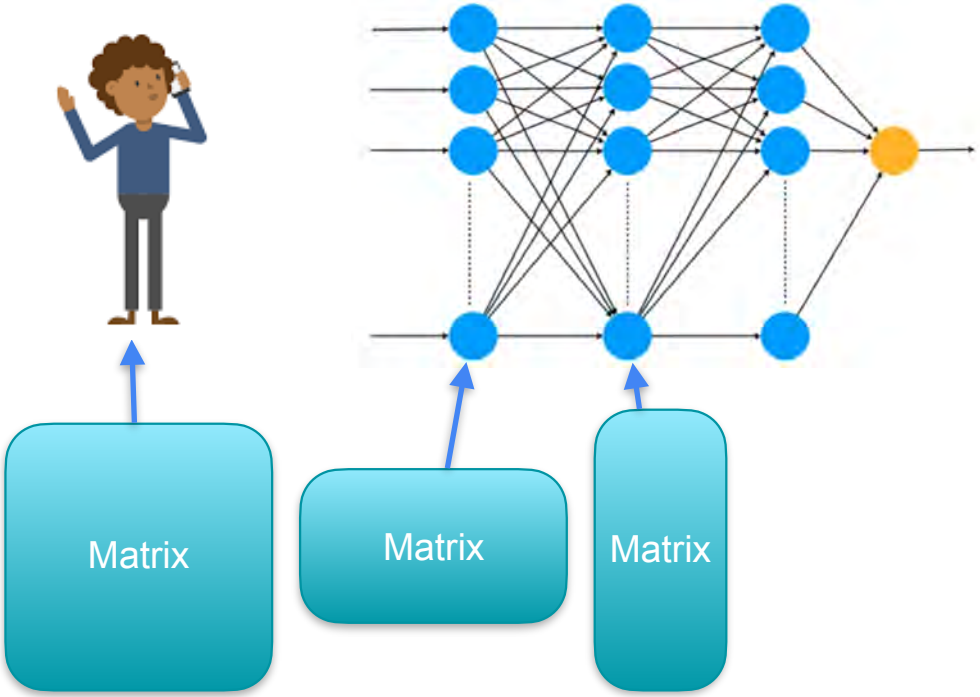
Neural networks - Matrix operations



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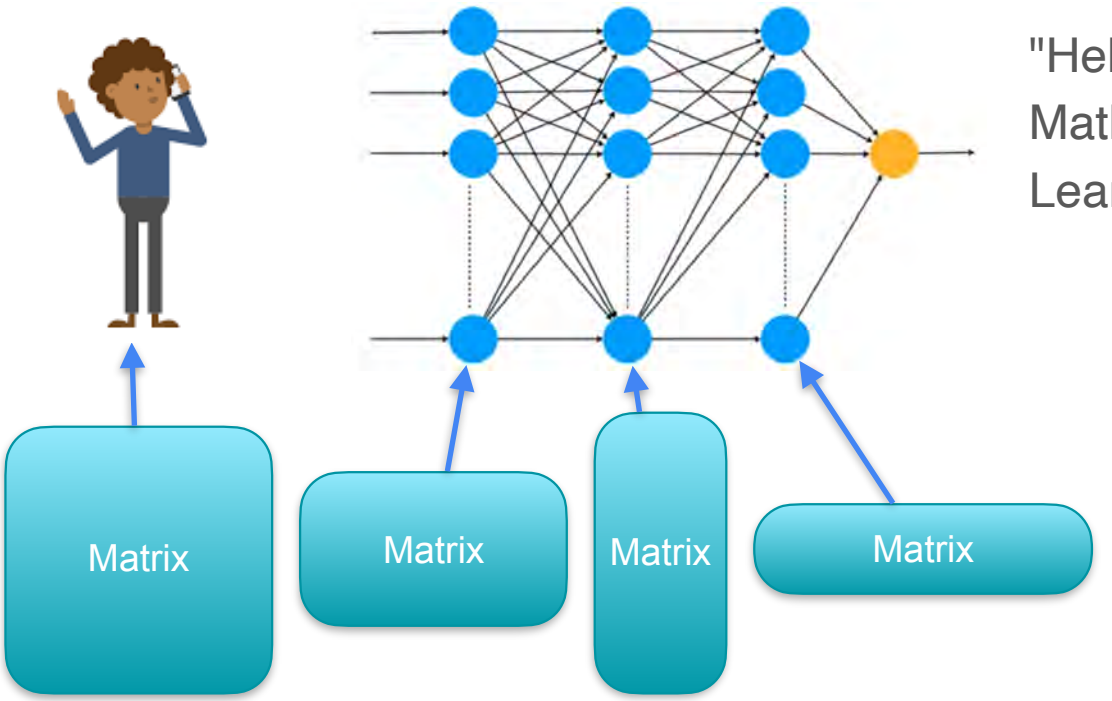
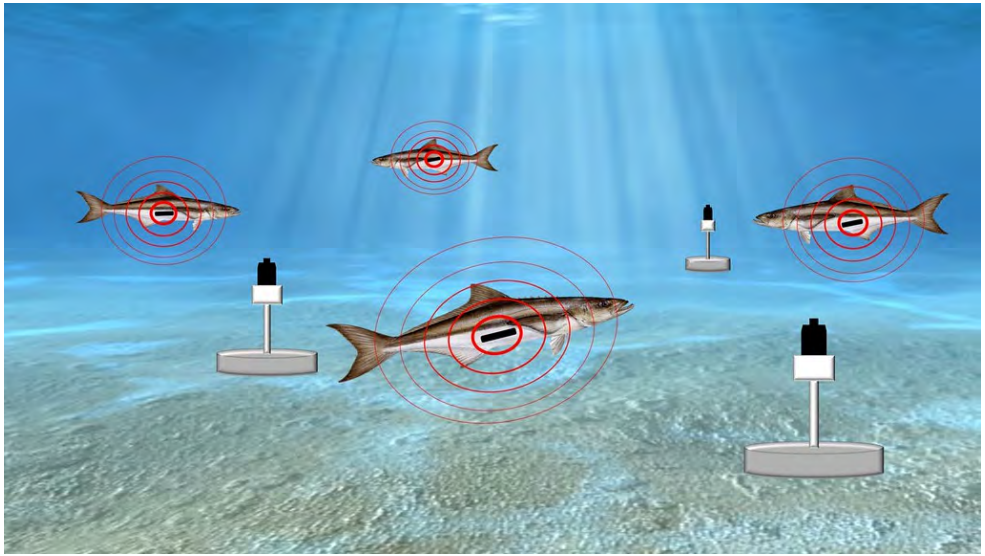
Neural networks - Matrix operations



"Hello! Welcome to Math for Machine Learning!"

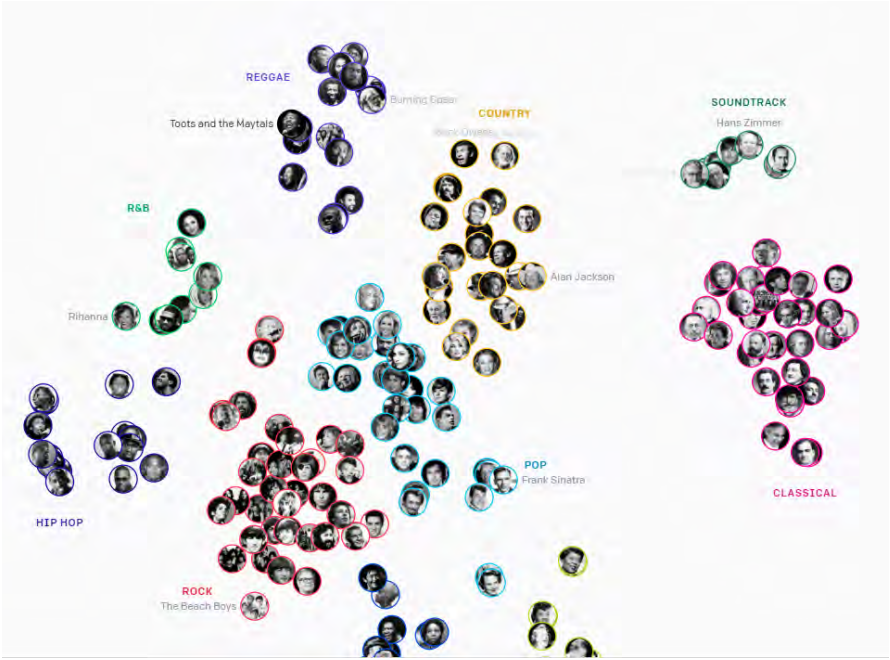
Neural networks - Sound recognition



Acoustic monitoring: Monitoring ecosystems through sounds

- Sound recognition: tracking species through sound to preserve bio-habitats.

Neural Networks - AI-generated music



Neural network generates music

- Automatic music generation: compressing music to discrete codes, then training the model on a specific genre to produce new music.

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Solving System of Linear Equations

Solving non-singular system of linear equations

Solving systems of equations

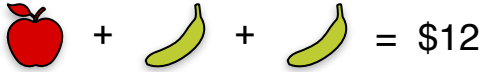
System

- $a + b = 10$
- $a + 2b = 12$

Solving systems of equations

System

- a + b = 10
- a + 2b = 12



Solving systems of equations

System

- a + b = 10
- a + 2b = 12

Solving systems of equations

System

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Solving systems of equations

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- a + b = 10
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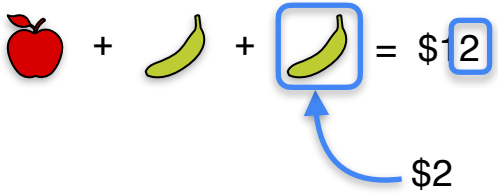
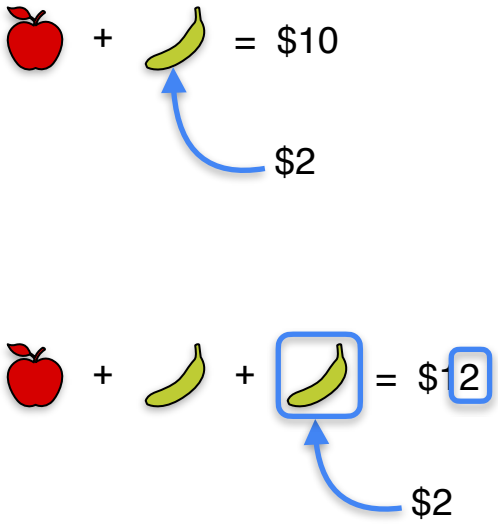


Solving systems of equations

System

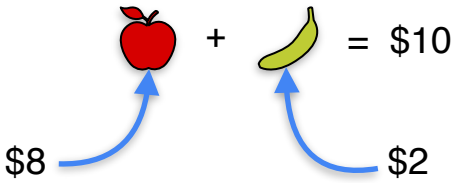
- a + b = 10
- a + 2b = 12

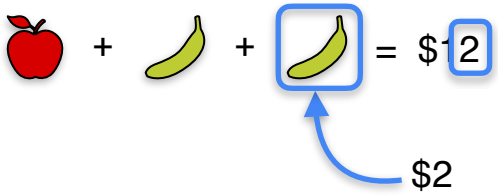


Solving systems of equations

System

- a + b = 10
- a + 2b = 12





Solving systems of equations

Solving systems of equations

System

- a + b = 10
- a + 2b = 12

Solving systems of equations

System

- $a + b = 10$
- $a + 2b = 12$

Solved system

- $a = 8$
- $b = 2$

Solving systems of equations

System

- $a + b = 10$
- $a + 2b = 12$

Some process →

Solved system

- $a = 8$
- $b = 2$

Solving systems of equations

System

- $a + b = 10$
- $a + 2b = 12$

Some process

Manipulating equations

Solved system

- $a = 8$
- $b = 2$

Solving systems of equations

System

- $a + b = 10$
- $a + 2b = 12$

Some process

Manipulating equations

Swapping equations
Adding equations
Multiplying equations by a constant

Solved system

- $a = 8$
- $b = 2$

Solving systems of equations

System

- $a + b = 10$
- $a + 2b = 12$

Solved system

- $a = 8$
- $b = 2$

Solving systems of equations

System

- $a + b = 10$
- $a + 2b = 12$

Eliminate ‘a’ from this equation

Solved system

- $a = 8$
- $b = 2$

Manipulating equations

Manipulating equations

Multiplying by a constant

Manipulating equations

Multiplying by a constant

$$a + b = 10$$

Manipulating equations

Multiplying by a constant

$$a + b = 10$$

$$\times \quad 7$$

Manipulating equations

Multiplying by a constant

$$a + b = 10$$

$$\times \quad 7$$

$$7a + 7b = 70$$

Manipulating equations

Multiplying by a constant

$$a + b = 10$$

$$\times \quad 7$$

$$7a + 7b = 70$$

Adding two equations

Manipulating equations

Multiplying by a constant

$$a + b = 10$$

$$\times \quad 7$$

$$7a + 7b = 70$$

Adding two equations

$$a + b = 10$$

Manipulating equations

Multiplying by a constant

$$a + b = 10$$

$$\times \quad 7$$

$$7a + 7b = 70$$

Adding two equations

$$a + b = 10$$

$$2a + 3b = 26$$

Manipulating equations

Multiplying by a constant

$$\begin{array}{rr} & a + b = 10 \\ \times & 7 \\ \hline & 7a + 7b = 70 \end{array}$$

Adding two equations

$$\begin{array}{rr} & a + b = 10 \\ + & 2a + 3b = 26 \\ \hline \end{array}$$

Manipulating equations

Multiplying by a constant

$$\begin{array}{rr} & a + b = 10 \\ \times & 7 \\ \hline & 7a + 7b = 70 \end{array}$$

Adding two equations

$$\begin{array}{rr} & a + b = 10 \\ + & 2a + 3b = 26 \\ \hline & 3a + 4b = 36 \end{array}$$

Let’s do a harder example

Systems of equations

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Solved system

- $a = ?$
- $b = ?$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Eliminate ‘a’
from this equation

Solved system

- $a = ?$
- $b = ?$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Eliminate ‘a’
from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- $a - 0.75b = 1.5$

Solved system

- $a = ?$
- $b = ?$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
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Subtract equation 1 from equation 2

Solved system

- $a = ?$
- $b = ?$

Systems of equations

System

- $5a + b = 17$
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Subtract equation 1 from equation 2

$a - 0.75b = 1.5$

Solved system

- $a = ?$
- $b = ?$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Eliminate ‘a’
from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- $a - 0.75b = 1.5$

Subtract equation 1 from equation 2

$$\begin{array}{rl} & a - 0.75b = 1.5 \\ - & a + 0.2b = 3.4 \\ \hline \end{array}$$

Solved system

- $a = ?$
- $b = ?$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Eliminate ‘a’
from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- $a - 0.75b = 1.5$

Subtract equation 1 from equation 2

$$
\begin{array}{rl}
 & a - 0.75b = 1.5 \\
- & a + 0.2b = 3.4 \\
\hline
 & 0a - 0.95b = -1.9
\end{array}
$$

Solved system

- $a = ?$
- $b = ?$

Systems of equations

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- $5a + b = 17$
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- & a + 0.2b = 3.4 \\
\hline
 & 0a - 0.95b = -1.9 \\
 & -0.95b = -1.9 \\
 & b = 2
\end{array}
$$

Solved system

- $a = ?$
- $b = ?$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

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Solved system

- $a = ?$
- $b = 2$

Systems of equations

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Eliminate ‘a’ from this equation

Divide by coefficient of a

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Subtract equation 1 from equation 2

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- & a + 0.2b = 3.4 \\
\hline
 & 0a - 0.95b = -1.9 \\
 & -0.95b = -1.9 \\
 & b = 2
\end{array}
$$

Solved system

- $a = ?$
- $b = 2$

$a + 0.2(2) = 3.4$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Eliminate ‘a’
from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- $a - 0.75b = 1.5$

Subtract equation 1 from equation 2

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\begin{array}{rl}
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- & a + 0.2b = 3.4 \\
\hline
 & 0a - 0.95b = -1.9 \\
 & -0.95b = -1.9 \\
 & b = 2
\end{array}
$$

Solved system

- $a = ?$
- $b = 2$

$a + 0.2(2) = 3.4$

$a + 0.4 = 3.4$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

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Subtract equation 1 from equation 2

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\begin{array}{rl}
 & a - 0.75b = 1.5 \\
- & a + 0.2b = 3.4 \\
\hline
 & 0a - 0.95b = -1.9 \\
 & -0.95b = -1.9 \\
 & b = 2
\end{array}
$$

Solved system

- $a = ?$
- $b = 2$

$$
\begin{aligned}
a + 0.2(2) &= 3.4 \\
a + 0.4 &= 3.4 \\
a &= 3
\end{aligned}
$$

Systems of equations

System

- $5a + b = 17$
- $4a - 3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- $a - 0.75b = 1.5$

Subtract equation 1 from equation 2

$$
\begin{array}{rl}
 & a - 0.75b = 1.5 \\
- & a + 0.2b = 3.4 \\
\hline
 & 0a - 0.95b = -1.9 \\
 & -0.95b = -1.9 \\
 & b = 2
\end{array}
$$

Solved system

- $a = 3$
- $b = 2$

$$
\begin{aligned}
a + 0.2(2) &= 3.4 \\
a + 0.4 &= 3.4 \\
a &= 3
\end{aligned}
$$

What if one of the coefficients of a is zero?

System

- $5a + b = 17$
- $3b = 6$

Solved system

- $a = ?$
- $b = ?$

What if one of the coefficients of a is zero?

System

- $5a + b = 17$
- $3b = 6$

Eliminate ‘a’
from this equation

Solved system

- $a = ?$
- $b = ?$

What if one of the coefficients of a is zero?

System

- $5a + b = 17$
- $3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- ???

Solved system

- $a = ?$
- $b = ?$

What if one of the coefficients of a is zero?

System

- $5a + b = 17$
- $3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- ???

$b = 2$

Solved system

- $a = ?$
- $b = ?$

What if one of the coefficients of a is zero?

System

- $5a + b = 17$
- $3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- ???

$b = 2$

Solved system

- $a = ?$
- $b = 2$

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- $3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- ???

$b = 2$

Solved system

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- $b = 2$

$a + 0.2(2) = 3.4$

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- $5a + b = 17$
- $3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- ???

$b = 2$

Solved system

- $a = ?$
- $b = 2$

$a + 0.2(2) = 3.4$

$a + 0.4 = 3.4$

What if one of the coefficients of a is zero?

System

- $5a + b = 17$
- $3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- ???

$b = 2$

Solved system

- $a = ?$
- $b = 2$

$a + 0.2(2) = 3.4$

$a + 0.4 = 3.4$

$a = 3$

What if one of the coefficients of a is zero?

System

- $5a + b = 17$
- $3b = 6$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + 0.2b = 3.4$
- ???

$b = 2$

Solved system

- $a = 3$
- $b = 2$

$a + 0.2(2) = 3.4$

$a + 0.4 = 3.4$

$a = 3$

Quiz

- Solve the following system of equations

System

- $2a + 5b = 46$
- $8a + b = 32$

Solution

- Solve the following system of equations

System

- $2a + 5b = 46$
- $8a + b = 32$

Solution

- $a = 3$
- $b = 8$

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Solving System of Linear Equations

Solving singular system of linear equations

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 10$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 10$

Subtract equation 1 from equation 2

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 10$

Subtract equation 1 from equation 2

$a + b = 10$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 10$

Subtract equation 1 from equation 2

$$\begin{array}{rl} & a + b = 10 \\ - & a + b = 10 \\ \hline \end{array}$$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 10$

Subtract equation 1 from equation 2

$$\begin{array}{rl} & a + b = 10 \\ - & a + b = 10 \\ \hline & 0 = 0 \end{array}$$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Eliminate 'a' from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 10$

Subtract equation 1 from equation 2

$$\begin{array}{rl} & a + b = 10 \\ - & a + b = 10 \\ \hline & 0 = 0 \end{array}$$

Solved system

- $a = ?$
- $b = ?$

Solved system

- $a + b = 10$
- no other equation

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Eliminate 'a' from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 10$

Subtract equation 1 from equation 2

$$\begin{array}{rl} & a + b = 10 \\ - & a + b = 10 \\ \hline & 0 = 0 \end{array}$$

Solved system

- $a = x$
- $b = ?$

Solved system

- $a + b = 10$
- no other equation

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Eliminate 'a' from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 10$

Subtract equation 1 from equation 2

$$\begin{array}{rl} & a + b = 10 \\ - & a + b = 10 \\ \hline & 0 = 0 \end{array}$$

Solved system

- $a = x$
- $b = 10 - x$

Solved system

- $a + b = 10$
- no other equation

What if the system is singular (redundant)?

System

- $a + b = 10$
- $2a + 2b = 20$

Eliminate 'a' from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 10$

Subtract equation 1 from equation 2

$$\begin{array}{rl} & a + b = 10 \\ - & a + b = 10 \\ \hline & 0 = 0 \end{array}$$

Degree of freedom x

Solved system

- $a = x$
- $b = 10 - x$

Solved system

- $a + b = 10$
- no other equation

What if the system is singular (contradictory)?

System

- $a + b = 10$
- $2a + 2b = 24$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (contradictory)?

System

- $a + b = 10$
- $2a + 2b = 24$

Eliminate ‘a’
from this equation

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (contradictory)?

System

- $a + b = 10$
- $2a + 2b = 24$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 12$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (contradictory)?

System

- $a + b = 10$
- $2a + 2b = 24$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 12$

Subtract equation 1 from equation 2

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (contradictory)?

System

- $a + b = 10$
- $2a + 2b = 24$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 12$

Subtract equation 1 from equation 2

$a + b = 12$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (contradictory)?

System

- $a + b = 10$
- $2a + 2b = 24$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 12$

Subtract equation 1 from equation 2

$$\begin{array}{rl} & a + b = 12 \\ - & a + b = 10 \\ \hline \end{array}$$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (contradictory)?

System

- $a + b = 10$
- $2a + 2b = 24$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 12$

Subtract equation 1 from equation 2

$$
\begin{array}{rl}
 & a + b = 12 \\
- & a + b = 10 \\
\hline
 & 0 = 2
\end{array}
$$

Solved system

- $a = ?$
- $b = ?$

What if the system is singular (contradictory)?

System

- $a + b = 10$
- $2a + 2b = 24$

Eliminate ‘a’ from this equation

Divide by coefficient of a

- $a + b = 10$
- $a + b = 12$

Subtract equation 1 from equation 2

$$
\begin{array}{rl}
 & a + b = 12 \\
- & a + b = 10 \\
\hline
 & 0 = 2
\end{array}
$$

Contradiction!

Solved system

- $a = ?$
- $b = ?$

Quiz

- Solve the following system of equations

System

- $5a + b = 11$
- $10a + 2b = 22$

Solution

- Solve the following system of equations

System

- $5a + b = 11$
- $10a + 2b = 22$

Solution: If you look closely into the two equations in the system, you'll find that if equation 2 is divided by 2 you'll obtain equation 1.

Therefore, the system has infinitely many solutions.

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Solving System of Linear Equations

Solving system of equations with more variables

Elimination method

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Elimination method

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Leave ‘a’ by itself

Elimination method

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Elimination method

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Divide each row by the coefficient of ‘a’

Elimination method

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Divide each row by the coefficient of ‘a’

- $a + b + 2c = 12$
- $a - b - 1/3\ c = 1$
- $a - b/2 + 3c = 12$

Elimination method

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Divide each row by the coefficient of ‘a’

- $a + b + 2c = 12$
- $a - b - 1/3\ c = 1$
- $a - b/2 + 3c = 12$

Use the first equation to remove ‘a’ from the others

Elimination method

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Divide each row by the coefficient of ‘a’

- $a + b + 2c = 12$
- $a - b - 1/3\ c = 1$
- $a - b/2 + 3c = 12$

Use the first equation to remove ‘a’ from the others

- $a + b + 2c = 12$
- $-2b - 7/3\ c = -11$
- $-3/2\ b + c = 0$

Elimination method

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Divide each row by the coefficient of ‘a’

- $a + b + 2c = 12$
- $a - b - 1/3\ c = 1$
- $a - b/2 + 3c = 12$

Use the first equation to remove ‘a’ from the others

- $a + b + 2c = 12$
- $-2b - 7/3\ c = -11$
- $-3/2\ b + c = 0$

Isolated ‘a’

Elimination method

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Divide each row by the coefficient of ‘a’

- $a + b + 2c = 12$
- $a - b - 1/3\ c = 1$
- $a - b/2 + 3c = 12$

Use the first equation to remove ‘a’ from the others

- $a + b + 2c = 12$
- $-2b - 7/3\ c = -11$
- $-3/2\ b + c = 0$

Isolated ‘a’

Solve this new system of 2 equations

Elimination method

System

- $a + b + 2c = \mathbf{12}$
- $-2b - 7/3\ c = \mathbf{-11}$
- $-3/2\ b + c = \mathbf{0}$

Elimination method

System

- $a + b + 2c = 12$
- $-2b - 7/3\ c = -11$
- $-3/2\ b + c = 0$

Divide last two rows by the coefficient of b

Elimination method

System

- $a + b + 2c = 12$
- $-2b - 7/3\ c = -11$
- $-3/2\ b + c = 0$

Divide last two rows by the coefficient of b

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $b - 2/3\ c = 0$

Elimination method

System

- $a + b + 2c = 12$
- $-2b - 7/3\ c = -11$
- $-3/2\ b + c = 0$

Divide last two rows by the coefficient of b

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $b - 2/3\ c = 0$

Use the second equation to remove ‘b’ from the third

Elimination method

System

- $a + b + 2c = 12$
- $-2b - 7/3\ c = -11$
- $-3/2\ b + c = 0$

Divide last two rows by the coefficient of b

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $b - 2/3\ c = 0$

Use the second equation to remove ‘b’ from the third

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $-11/6\ c = -11/2$

Elimination method

System

- $a + b + 2c = 12$
- $-2b - 7/3\ c = -11$
- $-3/2\ b + c = 0$

Divide last two rows by the coefficient of b

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $b - 2/3\ c = 0$

Use the second equation to remove ‘b’ from the third

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $-11/6\ c = -11/2$

Isolated ‘b’

Elimination method

System

- $a + b + 2c = 12$
- $-2b - 7/3\ c = -11$
- $-3/2\ b + c = 0$

Divide last two rows by the coefficient of b

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $b - 2/3\ c = 0$

Use the second equation to remove ‘b’ from the third

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $-11/6\ c = -11/2$

Isolated ‘b’

c = 3

Elimination method

System

- $a + b + 2c = \mathbf{12}$
- $b + 7/6\ c = \mathbf{11/2}$
- $c = \mathbf{3}$

Elimination method

System

- $a + b + 2c = \mathbf{12}$
- $b + 7/6\ c = \mathbf{11/2}$
- $c = \mathbf{3}$

Replace c = 3 in the second equation, get b = 2

Elimination method

System

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$ → $b + 7/2 = 11/2$, $b = 2$
- $c = 3$

Replace c = 3 in the second equation, get b = 2

Elimination method

System

- $a + b + 2c = \mathbf{12}$
- $b + 7/6\ c = \mathbf{11/2}$ → $b + 7/2 = \mathbf{11/2}$, $b = \mathbf{2}$
- $c = \mathbf{3}$

Replace c = 3 in the second equation, get b = 2

Replace c = 3 and b = 2 in the first equation, get a = 4

Elimination method

System

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $c = 3$

$a + 2 + 6 = 12$
$a = 4$

$b + 7/2 = 11/2$
$b = 2$

Replace c = 3 in the second equation, get b = 2

Replace c = 3 and b = 2 in the first equation, get a = 4

Elimination method

System

- $a + b + 2c = 12$
- $b + 7/6\ c = 11/2$
- $c = 3$

$a + 2 + 6 = 12$
$a = 4$

$b + 7/2 = 11/2$
$b = 2$

Replace c = 3 in the second equation, get b = 2

Replace c = 3 and b = 2 in the first equation, get a = 4

The solution is
a = 4
b = 2
c = 3

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Solving System of Linear Equations

Matrix row reduction

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Original matrix

5	1
4	-3

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Diagonal matrix

1	0
0	1

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Diagonal matrix

1	0
0	1

Row echelon form

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $1a + 0b = 3$
- $0a + 1b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Row echelon form

Diagonal matrix

1	0
0	1

Reduced row echelon form

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $1a + 0b = 3$
- $0a + 1b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Row echelon form

Diagonal matrix

1	0
0	1

Reduced row echelon form

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $1a + 0b = 3$
- $0a + 1b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Row echelon form

Diagonal matrix

1	0
0	1

Reduced row echelon form

Systems of equations to matrices

Original system

- $a + b = 10$
- $2a + 2b = 20$

Systems of equations to matrices

Original system

- $a + b = 10$
- $2a + 2b = 20$

Intermediate System

- $a + b = 10$
- $0a + 0b = 0$

Systems of equations to matrices

Original system

- $a + b = 10$
- $2a + 2b = 20$

Intermediate System

- $a + b = 10$
- $0a + 0b = 0$

Original matrix

1	1
2	2

Systems of equations to matrices

Original system

- $a + b = 10$
- $2a + 2b = 20$

Intermediate System

- $a + b = 10$
- $0a + 0b = 0$

Original matrix

1	1
2	2

Upper diagonal matrix

1	1
0	0

Systems of equations to matrices

Original system

- $a + b = 10$
- $2a + 2b = 20$

Intermediate System

- $a + b = 10$
- $0a + 0b = 0$

Original matrix

1	1
2	2

Upper diagonal matrix

1	1
0	0

Row echelon form

Systems of equations to matrices

Original system

- $a + b = 10$
- $2a + 2b = 20$

Intermediate System

- $a + b = 10$
- $0a + 0b = 0$

Original matrix

1	1
2	2

Upper diagonal matrix

1	1
0	0

Row echelon form

Systems of equations to matrices

Original system

- $5a + b = 11$
- $10a + 2b = 22$

Systems of equations to matrices

Original system

- $5a + b = 11$
- $10a + 2b = 22$

Intermediate System

- $a + 0.2b = 2.2$
- $0a + 0b = 0$

Systems of equations to matrices

Original system

- $5a + b = 11$
- $10a + 2b = 22$

Intermediate System

- $a + 0.2b = 2.2$
- $0a + 0b = 0$

Original matrix

5	1
10	2

Systems of equations to matrices

Original system

- $5a + b = 11$
- $10a + 2b = 22$

Intermediate System

- $a + 0.2b = 2.2$
- $0a + 0b = 0$

Original matrix

5	1
10	2

Upper diagonal matrix

1	0.2
0	0

Systems of equations to matrices

Original system

- $5a + b = 11$
- $10a + 2b = 22$

Intermediate System

- $a + 0.2b = 2.2$
- $0a + 0b = 0$

Original matrix

5	1
10	2

Upper diagonal matrix

1	0.2
0	0

Row echelon form

Systems of equations to matrices

Original system

- $5a + b = 11$
- $10a + 2b = 22$

Intermediate System

- $a + 0.2b = 2.2$
- $0a + 0b = 0$

Original matrix

5	1
10	2

Upper diagonal matrix

1	0.2
0	0

Row echelon form

Systems of equations to matrices

Original system

- $0a + 0b = 0$
- $0a + 0b = 0$

Systems of equations to matrices

Original system

- $0a + 0b = 0$
- $0a + 0b = 0$

Intermediate System

- $0a + 0b = 0$
- $0a + 0b = 0$

Systems of equations to matrices

Original system

- $0a + 0b = 0$
- $0a + 0b = 0$

Intermediate System

- $0a + 0b = 0$
- $0a + 0b = 0$

Original matrix

0	0
0	0

Systems of equations to matrices

Original system

- $0a + 0b = 0$
- $0a + 0b = 0$

Intermediate System

- $0a + 0b = 0$
- $0a + 0b = 0$

Original matrix

0	0
0	0

Upper diagonal matrix

0	0
0	0

Systems of equations to matrices

Original system

- $0a + 0b = 0$
- $0a + 0b = 0$

Intermediate System

- $0a + 0b = 0$
- $0a + 0b = 0$

Original matrix

0	0
0	0

Upper diagonal matrix

0	0
0	0

Row echelon form

Systems of equations to matrices

Original system

- $0a + 0b = 0$
- $0a + 0b = 0$

Intermediate System

- $0a + 0b = 0$
- $0a + 0b = 0$

Original matrix

0	0
0	0

Upper diagonal matrix

0	0
0	0

Row echelon form

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Solving System of Linear Equations

Row operations that preserve singularity

Switching rows

5	1
4	3

Switching rows

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4 = 11$

Switching rows

5	1
4	3

4	3
5	1

Determinant $= 5 \cdot 3 - 1 \cdot 4 = 11$

Switching rows

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4 = 11$

4	3
5	1

Determinant $= 4 \cdot 1 - 3 \cdot 5 = -11$

Switching rows

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4 = 11$

4	3
5	1

Determinant $= 4 \cdot 1 - 3 \cdot 5 = -11$

Switching rows

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4 = 11$

4	3
5	1

Determinant $= 4 \cdot 1 - 3 \cdot 5 = -11$

Switching rows

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4 = 11$

4	3
5	1

Determinant $= 4 \cdot 1 - 3 \cdot 5 = -11$

Multiplying a row by a (non-zero) scalar

5	1
4	3

Multiplying a row by a (non-zero) scalar

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

Multiplying a row by a (non-zero) scalar

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Multiplying a row by a (non-zero) scalar

5	1
4	-3

4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Multiplying a row by a (non-zero) scalar

5	1
4	-3

5	1

4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Multiplying a row by a (non-zero) scalar

5	1
4	-3

5	1	x 10 =	
			4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Multiplying a row by a (non-zero) scalar

5	1
4	-3

5	1	x 10 =	50	10

4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Multiplying a row by a (non-zero) scalar

5	1
4	-3

5	1	x 10 =	50	10	

50	10
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Multiplying a row by a (non-zero) scalar

5	1
4	-3

5	1	x 10 =	50	10

50	10
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Determinant $= 5 \cdot (10 \cdot 3) - 1 \cdot (10 \cdot 4)$

Multiplying a row by a (non-zero) scalar

5	1
4	-3

5	1	x 10 =	50	10

50	10
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Determinant $= 5 \cdot (10 \cdot 3) - 1 \cdot (10 \cdot 4)$

$= 10 \cdot 11$

Adding a row to another row

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Adding a row to another row

5	1

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Adding a row to another row

5	1
4	3

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Adding a row to another row

5	1
4	3

$$\begin{array}{cc} & \begin{array}{cc} 5 & 1 \end{array} \\ + & \begin{array}{cc} 4 & 3 \end{array} \\ \hline \end{array}$$

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Adding a row to another row

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

$$
\begin{array}{cccc}
 & 5 & 1 \\
+ & 4 & 3 \\
\hline
 & 9 & 4
\end{array}
$$

Adding a row to another row

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

$$\begin{array}{cc} & \begin{array}{cc} 5 & 1 \end{array} \\ + & \begin{array}{cc} 4 & 3 \end{array} \\ \hline & \begin{array}{cc} 9 & 4 \end{array} \end{array}$$

9	4

Adding a row to another row

5	1
4	3

$$\begin{array}{r} \begin{array}{|c|c|} \hline 5 & 1 \\ \hline 4 & 3 \\ \hline \end{array} \\ + \\ \hline \begin{array}{|c|c|} \hline 9 & 4 \\ \hline \end{array} \end{array}$$

9	4
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Adding a row to another row

5	1
4	3

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

$$\begin{array}{r|c|c|} & 5 & 1 \\ + & 4 & 3 \\ \hline & 9 & 4 \end{array}$$

9	4
4	3

Determinant $= 9 \cdot 3 - 4 \cdot 4$

Adding a row to another row

$$\begin{bmatrix} 5 & 1 \\ 4 & 3 \end{bmatrix}$$

$$\begin{array}{rcc} & 5 & 1 \\ + & 4 & 3 \\ \hline & 9 & 4 \end{array}$$

$$\begin{bmatrix} 9 & 4 \\ 4 & 3 \end{bmatrix}$$

Determinant $= 5 \cdot 3 - 1 \cdot 4$

$= 11$

Determinant $= 9 \cdot 3 - 4 \cdot 4$

$= 11$

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Solving System of Linear Equations

Rank of a matrix

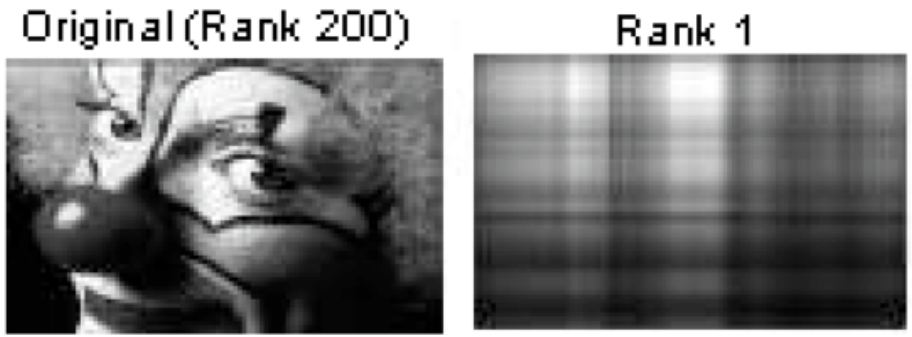
Compressing Images - Reducing rank

Compressing Images - Reducing rank

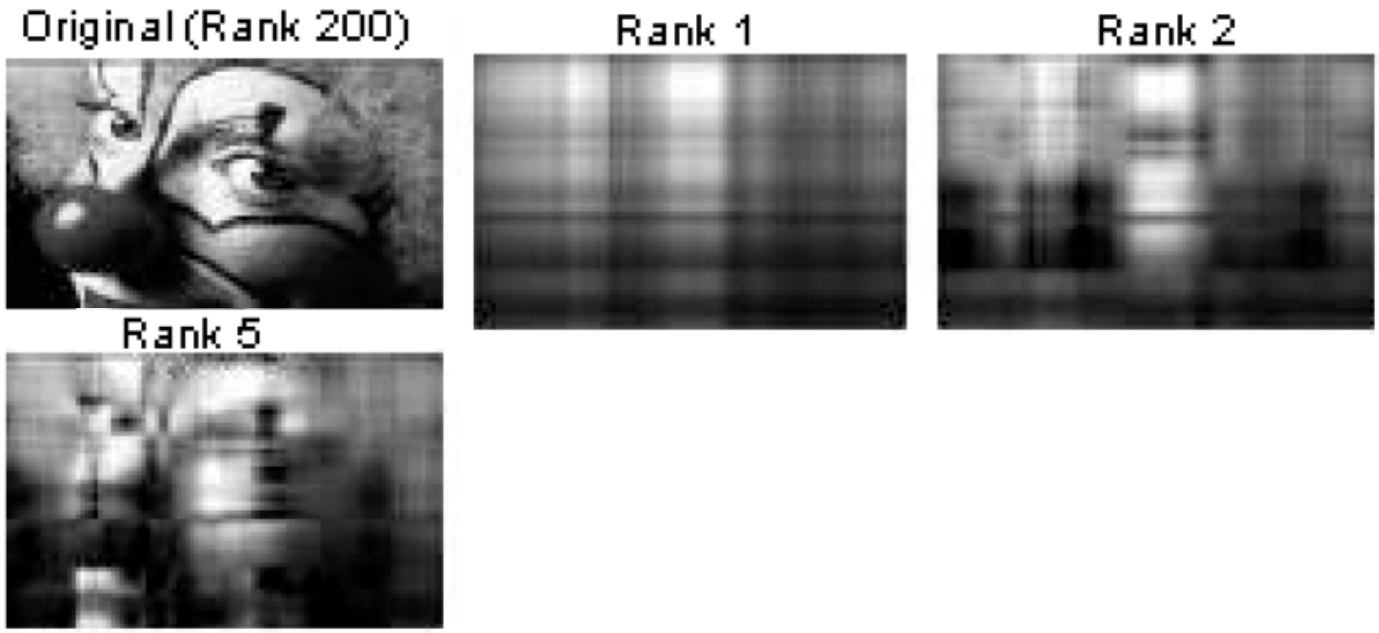
Compressing Images - Reducing rank

Original (Rank 200)

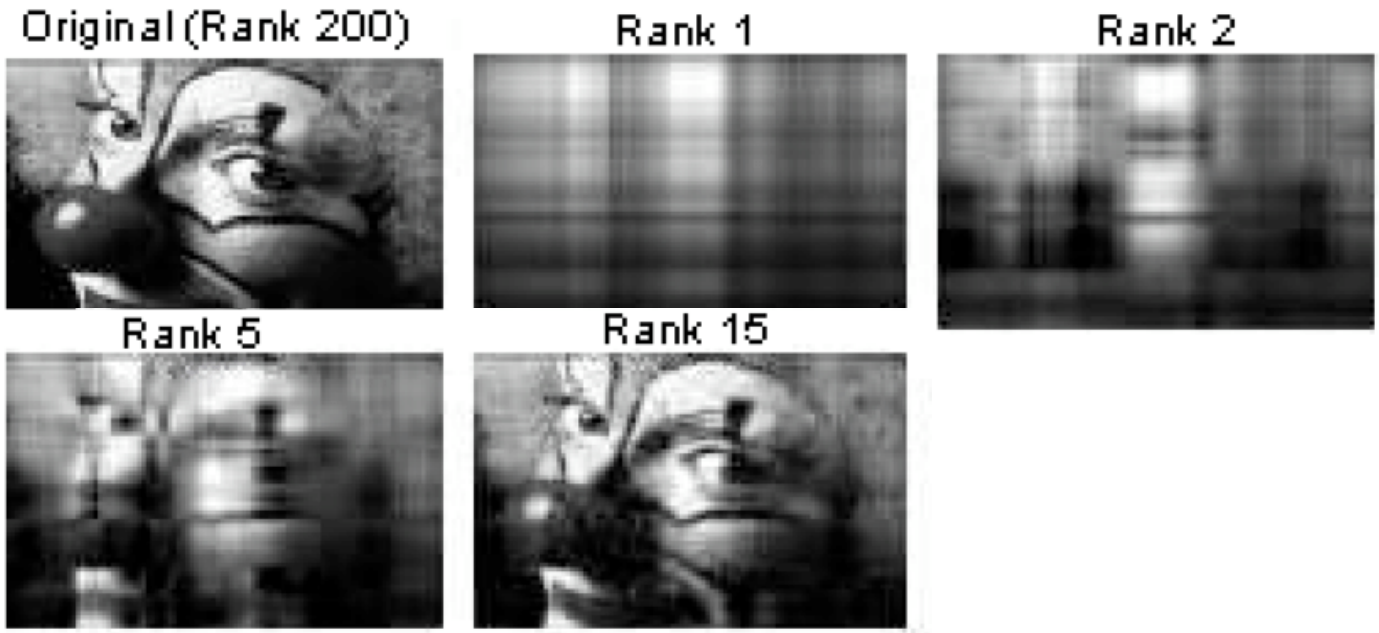
Compressing Images - Reducing rank



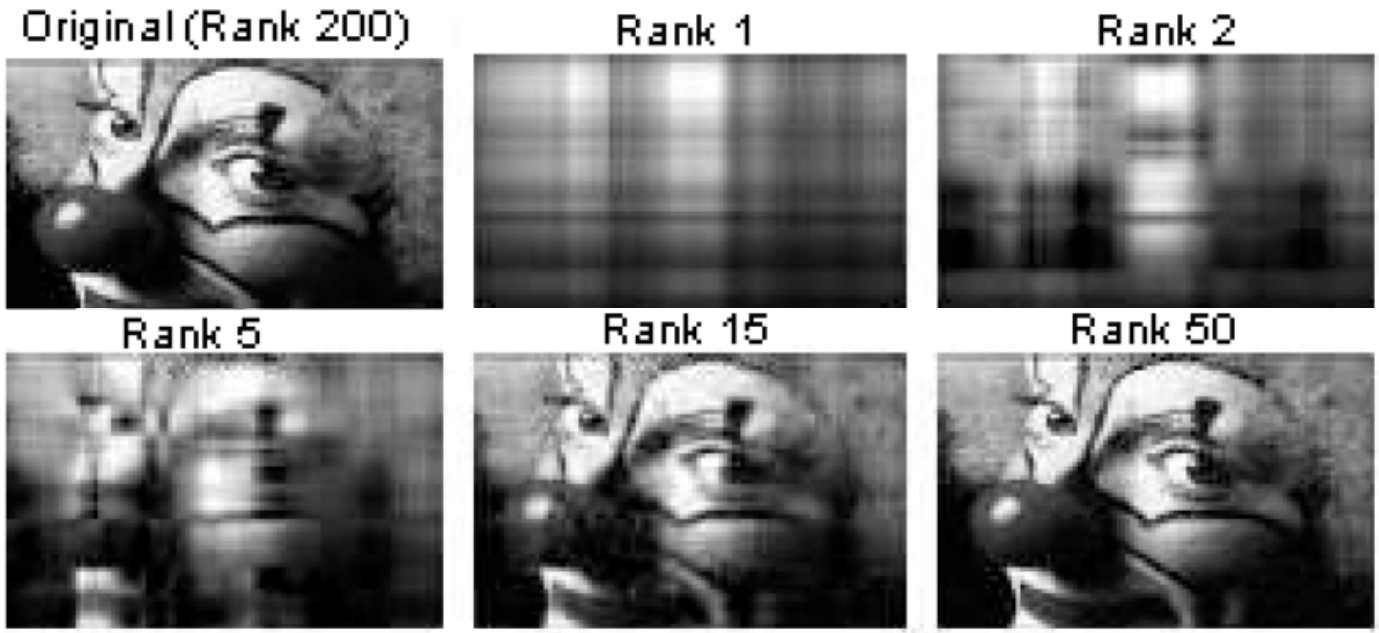
Compressing Images - Reducing rank



Compressing Images - Reducing rank



Compressing Images - Reducing rank



Systems of information

Systems of information

System 1

The dog is **black**

The cat is **orange**

Systems of information

System 1

The dog is **black**

The cat is **orange**

System 2

The dog is **black**

The dog is **black**

Systems of information

System 1

The dog is **black**

The cat is **orange**

System 2

The dog is **black**

The dog is **black**

System 3

The dog

The dog

Systems of information

System 1

The dog is **black**

The cat is **orange**

System 2

The dog is **black**

The dog is **black**

System 3

The dog

The dog

Two sentences

Systems of information

System 1

The dog is **black**

The cat is **orange**

System 2

The dog is **black**

The dog is **black**

System 3

The dog

The dog

Two sentences

Two pieces of information

Systems of information

System 1

The dog is **black**
The cat is **orange**

Two sentences

Two pieces of information

System 2

The dog is **black**
The dog is **black**

Two sentences

System 3

The dog
The dog

Systems of information

System 1

The dog is **black**
The cat is **orange**

Two sentences

Two pieces of information

System 2

The dog is **black**
The dog is **black**

Two sentences

One piece of information

System 3

The dog
The dog

Systems of information

System 1

The dog is **black**
The cat is **orange**

Two sentences

Two pieces of information

System 2

The dog is **black**
The dog is **black**

Two sentences

One piece of information

System 3

The dog
The dog

Two sentences

Systems of information

System 1

The dog is **black**
The cat is **orange**

Two sentences

Two pieces of information

System 2

The dog is **black**
The dog is **black**

Two sentences

One piece of information

System 3

The dog
The dog

Two sentences

Zero pieces of information

Systems of information

System 1

The dog is **black**
The cat is **orange**

Two sentences

Two pieces of information

Rank = 2

System 2

The dog is **black**
The dog is **black**

Two sentences

One piece of information

System 3

The dog
The dog

Two sentences

Zero pieces of information

Systems of information

System 1

The dog is **black**
The cat is **orange**

Two sentences

Two pieces of information

Rank = 2

System 2

The dog is **black**
The dog is **black**

Two sentences

One piece of information

Rank = 1

System 3

The dog
The dog

Two sentences

Zero pieces of information

Systems of information

System 1

The dog is **black**
The cat is **orange**

Two sentences

Two pieces of information

Rank = 2

System 2

The dog is **black**
The dog is **black**

Two sentences

One piece of information

Rank = 1

System 3

The dog
The dog

Two sentences

Zero pieces of information

Rank = 0

Systems of equations

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

System 2

$a + b = 0$

$2a + 2b = 0$

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

System 2

$a + b = 0$

$2a + 2b = 0$

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

System 2

$a + b = 0$

$2a + 2b = 0$

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

System 2

$a + b = 0$

$2a + 2b = 0$

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Two pieces of information

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

System 2

$a + b = 0$

$2a + 2b = 0$

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Two pieces of information

Rank = 2

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

Two equations

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

Two equations

One piece of information

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Zero pieces of information

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Zero pieces of information

Rank = 0

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

1	1
1	2

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Zero pieces of information

Rank = 0

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

1	1
1	2

Rank = 2

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Zero pieces of information

Rank = 0

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

1	1
1	2

Rank = 2

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

1	1
2	2

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Zero pieces of information

Rank = 0

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

1	1
1	2

Rank = 2

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

1	1
2	2

Rank = 1

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

Two equations

Zero pieces of information

Rank = 0

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

1	1
1	2

Rank = 2

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

1	1
2	2

Rank = 1

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

0	0
0	0

Two equations

Zero pieces of information

Rank = 0

Systems of equations

System 1

$a + b = 0$

$a + 2b = 0$

1	1
1	2

Rank = 2

Two equations

Two pieces of information

Rank = 2

System 2

$a + b = 0$

$2a + 2b = 0$

1	1
2	2

Rank = 1

Two equations

One piece of information

Rank = 1

System 3

$0a + 0b = 0$

$0a + 0b = 0$

0	0
0	0

Rank = 0

Two equations

Zero pieces of information

Rank = 0

Rank and solutions to the system

1	1
1	2

Rank = 2

1	1
2	2

Rank = 1

0	0
0	0

Rank = 0

Rank and solutions to the system

1	1
1	2

Rank = 2

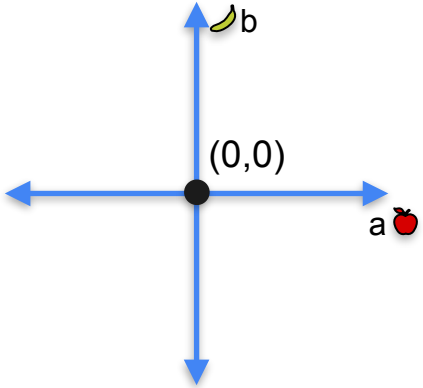
1	1
2	2

Rank = 1

0	0
0	0

Rank = 0

Dimension of solution space = 0



Rank and solutions to the system

1	1
1	2

Rank = 2

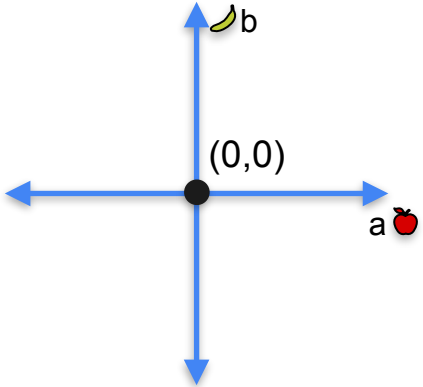
1	1
2	2

Rank = 1

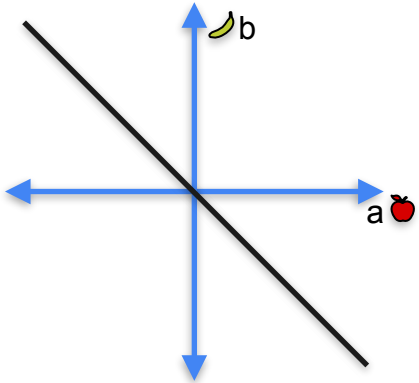
0	0
0	0

Rank = 0

Dimension of solution space = 0



Dimension of solution space = 1

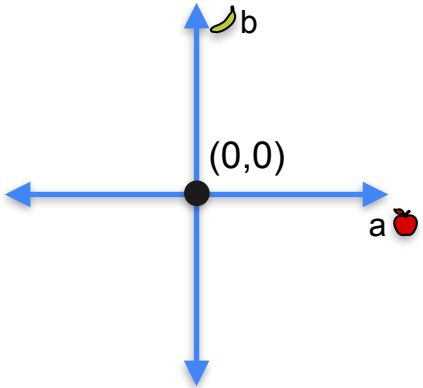


Rank and solutions to the system

1	1
1	2

Rank = 2

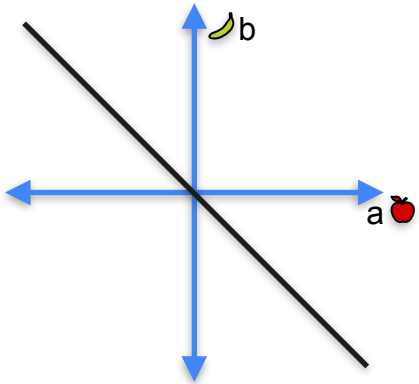
Dimension of solution space = 0



1	1
2	2

Rank = 1

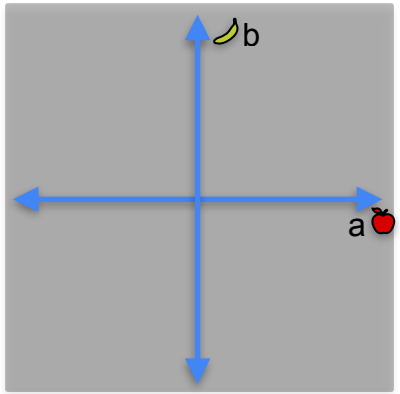
Dimension of solution space = 1



0	0
0	0

Rank = 0

Dimension of solution space = 2



Rank of a matrix

1	1
1	2

Rank = 2

Dimension of solution space = 0

1	1
2	2

Rank = 1

Dimension of solution space = 1

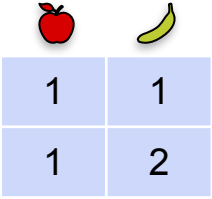
0	0
0	0

Rank = 0

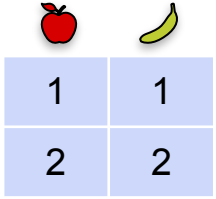
Dimension of solution space = 2

Rank = 2 - (Dimension of solution space)

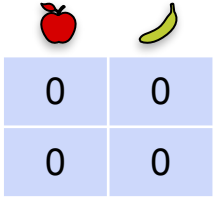
Rank and singularity



Rank = 2



Rank = 1



Rank = 0

Rank and singularity

1	1
1	2

Rank = 2

1	1
2	2

Rank = 1

0	0
0	0

Rank = 0

Non-singular

Rank and singularity

1	1
1	2

Rank = 2

Non-singular

1	1
2	2

Rank = 1

Singular

0	0
0	0

Rank = 0

Rank and singularity

1	1
1	2

Rank = 2

Non-singular

1	1
2	2

Rank = 1

Singular

0	0
0	0

Rank = 0

Singular

Rank and singularity

1	1
1	2

Rank = 2

Non-singular

1	1
2	2

Rank = 1

Singular

0	0
0	0

Rank = 0

Singular

Quiz: Rank of a matrix

Determine the rank of the following two matrices

Matrix 1

5	1
-1	3

Matrix 2

2	-1
-6	3

Solutions: Rank of a matrix

Determine the rank of the following two matrices

Matrix 1: Since the solution space had dimension 0, the rank is **2**.

5	1
-1	3

Matrix 2: Since the solution space had dimension 1, the rank is **1**.

2	-1
-6	3

DeepLearning.AI

Solving System of Linear Equations

Rank of a matrix: General case

Rank for matrices

System 1

$$a + b + c = 0$$
$$a + 2b + c = 0$$
$$a + b + 2c = 0$$

System 2

$$a + b + c = 0$$
$$a + b + 2c = 0$$
$$a + b + 3c = 0$$

System 3

$$a + b + c = 0$$
$$2a + 2b + 2c = 0$$
$$3a + 3b + 3c = 0$$

System 4

$$0a + 0b + 0c = 0$$
$$0a + 0b + 0c = 0$$
$$0a + 0b + 0c = 0$$

Rank for matrices

System 1

$$a + b + c = 0$$
$$a + 2b + c = 0$$
$$a + b + 2c = 0$$

System 2

$$a + b + c = 0$$
$$a + b + 2c = 0$$
$$a + b + 3c = 0$$

System 3

$$a + b + c = 0$$
$$2a + 2b + 2c = 0$$
$$3a + 3b + 3c = 0$$

System 4

$$0a + 0b + 0c = 0$$
$$0a + 0b + 0c = 0$$
$$0a + 0b + 0c = 0$$

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$

System 2

$a + b + c = 0$
$a + b + 2c = 0$
$a + b + 3c = 0$

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

System 2

$a + b + c = 0$
$a + b + 2c = 0$
$a + b + 3c = 0$

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank for matrices

System 1

$a + b + c = 0$ ✔
$a + 2b + c = 0$ ✔
$a + b + 2c = 0$ ✔

System 2

$a + b + c = 0$
$a + b + 2c = 0$
$a + b + 3c = 0$

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

3 Equations
3 Pieces of information

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

System 2

$a + b + c = 0$
$a + b + 2c = 0$
$a + b + 3c = 0$

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

3 Equations
3 Pieces of information

Rank 3

Rank for matrices

System 1

$a + b + c = 0$ ✔
$a + 2b + c = 0$ ✔
$a + b + 2c = 0$ ✔

System 2

$a + b + c = 0$
$a + b + 2c = 0$
$a + b + 3c = 0$

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$
$a + b + 3c = 0$

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$
$a + b + 3c = 0$ ✓

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank 3

1	1	1
1	2	1
1	1	2

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank 3

1	1	1
1	2	1
1	1	2

Rank 2

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

System 3

$a + b + c = 0$
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank 3

1	1	1
1	2	1
1	1	2

Rank 2

1	1	1
1	1	2
1	1	3

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank 3

1	1	1
1	2	1
1	1	2

Rank 2

1	1	1
1	1	2
1	1	3

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank 3

1	1	1
1	2	1
1	1	2

Rank 2

1	1	1
1	1	2
1	1	3

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank 3

1	1	1
1	2	1
1	1	2

Rank 2

1	1	1
1	1	2
1	1	3

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

3 Equations
1 Piece of information

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank 3

1	1	1
1	2	1
1	1	2

Rank 2

1	1	1
1	1	2
1	1	3

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

3 Equations
1 Piece of information

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank 3

1	1	1
1	2	1
1	1	2

Rank 2

1	1	1
1	1	2
1	1	3

Rank 1

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

Rank 2

1	1	1
1	1	2
1	1	3

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

3 Equations
1 Piece of information

Rank 1

1	1	1
2	2	2
3	3	3

System 4

$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

Rank 2

1	1	1
1	1	2
1	1	3

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

3 Equations
1 Piece of information

Rank 1

1	1	1
2	2	2
3	3	3

System 4

$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$
$0a + 0b + 0c = 0$

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

Rank 2

1	1	1
1	1	2
1	1	3

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

3 Equations
1 Piece of information

Rank 1

1	1	1
2	2	2
3	3	3

System 4

$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

Rank 2

1	1	1
1	1	2
1	1	3

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

3 Equations
1 Piece of information

Rank 1

1	1	1
2	2	2
3	3	3

System 4

$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$ ✗

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

Rank 2

1	1	1
1	1	2
1	1	3

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

3 Equations
1 Piece of information

Rank 1

1	1	1
2	2	2
3	3	3

System 4

$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$ ✗

3 Equations
0 Pieces of information

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

Rank 2

1	1	1
1	1	2
1	1	3

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

3 Equations
1 Piece of information

Rank 1

1	1	1
2	2	2
3	3	3

System 4

$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$ ✗

3 Equations
0 Pieces of information

Rank 0

Rank for matrices

System 1

$a + b + c = 0$ ✓
$a + 2b + c = 0$ ✓
$a + b + 2c = 0$ ✓

3 Equations
3 Pieces of information

Rank 3

1	1	1
1	2	1
1	1	2

System 2

$a + b + c = 0$ ✓
$a + b + 2c = 0$ ✗
$a + b + 3c = 0$ ✓

3 Equations
2 Pieces of information

Rank 2

1	1	1
1	1	2
1	1	3

System 3

$a + b + c = 0$ ✓
$2a + 2b + 2c = 0$ ✗
$3a + 3b + 3c = 0$ ✗

3 Equations
1 Piece of information

Rank 1

1	1	1
2	2	2
3	3	3

System 4

$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$ ✗
$0a + 0b + 0c = 0$ ✗

3 Equations
0 Pieces of information

Rank 0

0	0	0
0	0	0
0	0	0

Question

- Is there an easier way to calculate the rank?
- Answer: Yes! As before, it is the number of ones in the diagonal of the reduced row echelon form of the matrix.

DeepLearning.AI

Solving System of Linear Equations

Row echelon form

Row echelon form of a matrix

Row echelon form of a matrix

Original matrix

5	1
4	-3

Row echelon form of a matrix

Original matrix

5	1
4	-3

Row echelon form

1	0.2
0	1

Row echelon form of a matrix

Original matrix

5	1
4	-3

Row echelon form

1	0.2
0	1

5	1
10	2

Row echelon form of a matrix

Original matrix → **Row echelon form**

5	1
4	-3

1	0.2
0	1

5	1
10	2

1	1
0	0

Row echelon form of a matrix

Original matrix → **Row echelon form**

5	1
4	-3

1	0.2
0	1

5	1
10	2

1	1
0	0

0	0
0	0

Row echelon form of a matrix

Original matrix			Row echelon form	
5	1	→	1	0.2
4	-3		0	1

5	1	→	1	1
10	2		0	0

0	0	→	0	0
0	0		0	0

Row echelon form

Original matrix

5	1
4	-3

Row echelon form

Original matrix

5	1
4	-3

Divide each row by
the leftmost coefficient

Row echelon form

Original matrix

5	1
4	-3

Divide each row by
the leftmost coefficient

Row echelon form

Original matrix

5	1
4	-3

$\longrightarrow$

1	0.2

Divide each row by
the leftmost coefficient

Row echelon form

Original matrix

5	1
4	-3

→

1	0.2
1	-0.75

Divide each row by
the leftmost coefficient

Row echelon form

Original matrix

5	1
4	-3

→

1	0.2
1	-0.75

→

1	0.2

Divide each row by
the leftmost coefficient

Row echelon form

Original matrix

5	1
4	-3

→

1	0.2
1	-0.75

→

1	0.2

Divide each row by
the leftmost coefficient

1	-0.75

Row echelon form

Original matrix

5	1
4	-3

→

1	0.2
1	-0.75

→

1	0.2

Divide each row by
the leftmost coefficient

1	-0.75
1	0.2

Row echelon form

Original matrix

5	1
4	-3

→

1	0.2
1	-0.75

→

1	0.2

Divide each row by
the leftmost coefficient

	1	-0.75
-	1	0.2

Row echelon form

Original matrix

5	1
4	-3

→

1	0.2
1	-0.75

→

1	0.2

Divide each row by
the leftmost coefficient

	1	-0.75
-	1	0.2
	0	-0.95

Row echelon form

Original matrix

5	1
4	-3

→

1	0.2
1	-0.75

→

1	0.2
0	-0.95

Divide each row by
the leftmost coefficient

	1	-0.75
-	1	0.2
	0	-0.95

Row echelon form

Original matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 1 & -0.75 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & -0.95 \end{bmatrix}$$

Divide each row by
the leftmost coefficient

$$\begin{array}{rcc} & 1 & -0.75 \\ - & 1 & 0.2 \\ \hline & 0 & -0.95 \end{array}$$

Divide the second row by
the leftmost non-zero coefficient

Row echelon form

Original matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 1 & -0.75 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & -0.95 \end{bmatrix} \longrightarrow$$

Divide each row by
the leftmost coefficient

$$\begin{array}{rcc} & 1 & -0.75 \\ - & 1 & 0.2 \\ \hline & 0 & -0.95 \end{array}$$

Divide the second row by
the leftmost non-zero coefficient

Row echelon form

Original matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 1 & -0.75 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & -0.95 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \end{bmatrix}$$

Divide each row by
the leftmost coefficient

$$\begin{array}{rcc} & 1 & -0.75 \\ - & 1 & 0.2 \\ \hline & 0 & -0.95 \end{array}$$

Divide the second row by
the leftmost non-zero coefficient

Row echelon form

Original matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 1 & -0.75 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & -0.95 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Divide each row by
the leftmost coefficient

$$\begin{array}{rcc} & 1 & -0.75 \\ - & 1 & 0.2 \\ \hline & 0 & -0.95 \end{array}$$

Divide the second row by
the leftmost non-zero coefficient

Row echelon form

Original matrix

5	1
4	-3

Divide each row by
the leftmost coefficient

1	0.2
1	-0.75

	1	-0.75
-	1	0.2
	0	-0.95

1	0.2
0	-0.95

Divide the second row by
the leftmost non-zero coefficient

Row echelon form

1	0.2
0	1

Row echelon form for singular matrices

Original matrix

5	1
10	2

Row echelon form for singular matrices

Original matrix

5	1
10	2

Divide each row by
the leftmost coefficient

Row echelon form for singular matrices

Original matrix

5	1
10	2

Divide each row by
the leftmost coefficient

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2

Divide each row by
the leftmost coefficient

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

Divide each row by
the leftmost coefficient

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

→

1	0.2

Divide each row by
the leftmost coefficient

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

→

1	0.2

Divide each row by
the leftmost coefficient

1	0.2

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

→

1	0.2

Divide each row by
the leftmost coefficient

1	0.2
1	0.2

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

→

1	0.2

Divide each row by
the leftmost coefficient

	1	0.2
-	1	0.2

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

→

1	0.2

Divide each row by
the leftmost coefficient

	1	0.2
-	1	0.2
	0	0

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

→

1	0.2
0	0

Divide each row by
the leftmost coefficient

	1	0.2
-	1	0.2
	0	0

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

→

1	0.2
0	0

Divide each row by
the leftmost coefficient

	1	0.2
-	1	0.2
	0	0

Divide the second row by
the leftmost non-zero coefficient

Row echelon form for singular matrices

Original matrix

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 1 & 0.2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix} \longrightarrow$$

Divide each row by
the leftmost coefficient

$$\begin{array}{rcc} & 1 & 0.2 \\ - & 1 & 0.2 \\ \hline & 0 & 0 \end{array}$$

Divide the second row by
the leftmost non-zero coefficient

Row echelon form for singular matrices

Original matrix

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 1 & 0.2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \end{bmatrix}$$

Divide each row by
the leftmost coefficient

$$\begin{array}{rcc} & 1 & 0.2 \\ - & 1 & 0.2 \\ \hline & 0 & 0 \end{array}$$

Divide the second row by
the leftmost non-zero coefficient

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

→

1	0.2
0	0

→

1	0.2
?	?

Divide each row by
the leftmost coefficient

	1	0.2
-	1	0.2
	0	0

Divide the second row by
the leftmost non-zero coefficient

Row echelon form for singular matrices

Original matrix

5	1
10	2

→

1	0.2
1	0.2

→

Row echelon form

1	0.2
0	0

→

1	0.2
?	?

Divide each row by
the leftmost coefficient

	1	0.2
-	1	0.2
	0	0

Divide the second row by
the leftmost non-zero coefficient

Row echelon form for singular matrices

Original matrix

0	0
0	0

Row echelon form for singular matrices

Original matrix

0	0
0	0

Divide each row by
the leftmost coefficient

Row echelon form for singular matrices

Original matrix

0	0
0	0

Divide each row by
the leftmost coefficient

Row echelon form for singular matrices

Original matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} ? & ? \end{bmatrix}$$

Divide each row by
the leftmost coefficient

Row echelon form for singular matrices

Original matrix

0	0
0	0

→

?	?
?	?

Divide each row by
the leftmost coefficient

Row echelon form for singular matrices

Row echelon form

Original matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix}$$

Divide each row by
the leftmost coefficient

Row echelon form, singularity, and rank

5	1
4	-3

5	1
10	2

0	0
0	0

Row echelon form, singularity, and rank

5	1
4	-3

$\longrightarrow$

1	0.2
0	1

5	1
10	2

0	0
0	0

Row echelon form, singularity, and rank

5	1
4	-3

$\longrightarrow$

1	0.2
0	1

5	1
10	2

$\longrightarrow$

1	0.2
0	0

0	0
0	0

Row echelon form, singularity, and rank

5	1
4	-3

$\longrightarrow$

1	0.2
0	1

5	1
10	2

$\longrightarrow$

1	0.2
0	0

0	0
0	0

$\longrightarrow$

0	0
0	0

Row echelon form, singularity, and rank

5	1
4	-3

→

1	0.2
0	1

2 ones in the diagonal

5	1
10	2

→

1	0.2
0	0

0	0
0	0

→

0	0
0	0

Row echelon form, singularity, and rank

5	1
4	-3

→

1	0.2
0	1

Rank 2

2 ones in the diagonal

5	1
10	2

→

1	0.2
0	0

0	0
0	0

→

0	0
0	0

Row echelon form, singularity, and rank

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Rank 2

2 ones in the diagonal

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$$

1 one in the diagonal

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Row echelon form, singularity, and rank

5	1
4	-3

→

1	0.2
0	1

Rank 2

2 ones in the diagonal

5	1
10	2

→

1	0.2
0	0

Rank 1

1 one in the diagonal

0	0
0	0

→

0	0
0	0

Row echelon form, singularity, and rank

5	1
4	-3

→

1	0.2
0	1

Rank 2

2 ones in the diagonal

5	1
10	2

→

1	0.2
0	0

Rank 1

1 one in the diagonal

0	0
0	0

→

0	0
0	0

0 ones in the diagonal

Row echelon form, singularity, and rank

5	1
4	-3

→

1	0.2
0	1

Rank 2

2 ones in the diagonal

5	1
10	2

→

1	0.2
0	0

Rank 1

1 one in the diagonal

0	0
0	0

→

0	0
0	0

Rank 0

0 ones in the diagonal

Row echelon form, singularity, and rank

Non-singular matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Rank 2

2 ones in the diagonal

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$$

Rank 1

1 one in the diagonal

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank 0

0 ones in the diagonal

Row echelon form, singularity, and rank

Non-singular matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Rank 2

2 ones in the diagonal

Singular matrix

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$$

Rank 1

1 one in the diagonal

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank 0

0 ones in the diagonal

Row echelon form, singularity, and rank

Non-singular matrix

$$\begin{bmatrix} 5 & 1 \\ 4 & -3 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix}$$

Rank 2

2 ones in the diagonal

Singular matrix

$$\begin{bmatrix} 5 & 1 \\ 10 & 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0.2 \\ 0 & 0 \end{bmatrix}$$

Rank 1

1 one in the diagonal

Singular matrix

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank 0

0 ones in the diagonal

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Solving System of Linear Equations

Row echelon form: General case

Row echelon form

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Row echelon form

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

System

- $a + b + 2c = 12$
- $-6b - 7c = -33$
- $6c = 18$

Row echelon form

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

Matrix

1	1	2
3	-3	-1
2	-1	6

System

- $a + b + 2c = 12$
- $-6b - 7c = -33$
- $6c = 18$

Row echelon form

System

- $a + b + 2c = 12$
- $3a - 3b - c = 3$
- $2a - b + 6c = 24$

System

- $a + b + 2c = 12$
- $-6b - 7c = -33$
- $6c = 18$

Matrix

1	1	2
3	-3	-1
2	-1	6

Row echelon form matrix

1	1	2
0	-6	7
0	0	6

Row echelon form

2	*	*	*	*
0	**1**	*	*	*
0	0	**3**	*	*
0	0	0	**-5**	*
0	0	0	0	**1**

3	*	*	*	*
0	0	**1**	*	*
0	0	0	**-4**	*
0	0	0	**0**	0
0	0	0	0	**0**

Row echelon form

2	*	*	*	*
0	1	*	*	*
0	0	3	*	*
0	0	0	-5	*
0	0	0	0	1

3	*	*	*	*
0	0	1	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

- Zero rows at the bottom

Row echelon form

2	*	*	*	*
0	1	*	*	*
0	0	3	*	*
0	0	0	-5	*
0	0	0	0	1

3	*	*	*	*
0	0	1	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

- Zero rows at the bottom
- Each row has a pivot (leftmost non-zero entry)

Row echelon form

2	*	*	*	*
0	1	*	*	*
0	0	3	*	*
0	0	0	-5	*
0	0	0	0	1

3	*	*	*	*
0	0	1	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

- Zero rows at the bottom
- Each row has a pivot (leftmost non-zero entry)
- Every pivot is to the right of the pivots on the rows above

Row echelon form

2	*	*	*	*
0	1	*	*	*
0	0	3	*	*
0	0	0	-5	*
0	0	0	0	1

Rank 5

3	*	*	*	*
0	0	1	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

Rank 3

- Zero rows at the bottom
- Each row has a pivot (leftmost non-zero entry)
- Every pivot is to the right of the pivots on the rows above
- Rank of the matrix is the number of pivots

Another example

Matrix

1	1	1
1	2	1
1	1	2

Another example

Matrix

1	1	1
1	2	1
1	1	2

Subtract the first row from the second and the third ones

Another example

Matrix

1	1	1
1	2	1
1	1	2

Subtract the first row from the second and the third ones

Row echelon form

1	1	1
0	1	0
0	0	1

What if the matrix is singular?

Matrix

1	1	1
1	1	2
1	1	3

What if the matrix is singular?

Matrix

1	1	1
1	1	2
1	1	3

Subtract the first row from the second and the third ones

What if the matrix is singular?

Matrix

1	1	1
1	1	2
1	1	3

1	1	1
0	0	1
0	0	2

Subtract the first row from the second and the third ones

What if the matrix is singular?

Matrix

1	1	1
1	1	2
1	1	3

Subtract the first row from the second and the third ones

1	1	1
0	0	1
0	0	2

Subtract twice the second row from the third one

What if the matrix is singular?

Matrix

1	1	1
1	1	2
1	1	3

Subtract the first row from the second and the third ones

1	1	1
0	0	1
0	0	2

Subtract twice the second row from the third one

Row echelon form

1	1	1
0	0	1
0	0	0

What if the matrix is singular?

Matrix

1	1	1
2	2	2
3	3	3

What if the matrix is singular?

Matrix

1	1	1
2	2	2
3	3	3

Subtract twice the first row from the second row

What if the matrix is singular?

Matrix

1	1	1
2	2	2
3	3	3

1	1	1
0	0	0
3	3	3

Subtract twice the first row from the second row

What if the matrix is singular?

Matrix

1	1	1
2	2	2
3	3	3

Subtract twice the first row from the second row

1	1	1
0	0	0
3	3	3

Subtract three times the first row from the third row

What if the matrix is singular?

Matrix

1	1	1
2	2	2
3	3	3

Subtract twice the first row from the second row

1	1	1
0	0	0
3	3	3

Subtract three times the first row from the third row

Row echelon form

1	1	1
0	0	0
0	0	0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

1	1	1
0	0	1
0	0	0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

1	1	1
0	0	1
0	0	0

1	1	1
0	0	0
0	0	0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

1	1	1
0	0	1
0	0	0

1	1	1
0	0	0
0	0	0

0	0	0
0	0	0
0	0	0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

1	1	1
0	0	1
0	0	0

1	1	1
0	0	0
0	0	0

0	0	0
0	0	0
0	0	0

Number of pivots = 3

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

Number of pivots = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

1	1	1
0	0	0
0	0	0

0	0	0
0	0	0
0	0	0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

Number of pivots = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

1	1	1
0	0	0
0	0	0

Number of pivots = 1

0	0	0
0	0	0
0	0	0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

Number of pivots = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

1	1	1
0	0	0
0	0	0

Number of pivots = 1

0	0	0
0	0	0
0	0	0

Number of pivots = 0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Rank = 3

Matrix 2

1	1	1
1	1	2
1	1	3

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

Number of pivots = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

1	1	1
0	0	0
0	0	0

Number of pivots = 1

0	0	0
0	0	0
0	0	0

Number of pivots = 0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Rank = 3

Matrix 2

1	1	1
1	1	2
1	1	3

Rank = 2

Matrix 3

1	1	1
2	2	2
3	3	3

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

Number of pivots = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

1	1	1
0	0	0
0	0	0

Number of pivots = 1

0	0	0
0	0	0
0	0	0

Number of pivots = 0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Rank = 3

Matrix 2

1	1	1
1	1	2
1	1	3

Rank = 2

Matrix 3

1	1	1
2	2	2
3	3	3

Rank = 1

Matrix 4

0	0	0
0	0	0
0	0	0

Row echelon forms

1	1	1
0	1	0
0	0	1

Number of pivots = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

1	1	1
0	0	0
0	0	0

Number of pivots = 1

0	0	0
0	0	0
0	0	0

Number of pivots = 0

Rank for matrices

Matrix 1

1	1	1
1	2	1
1	1	2

Rank = 3

Matrix 2

1	1	1
1	1	2
1	1	3

Rank = 2

Matrix 3

1	1	1
2	2	2
3	3	3

Rank = 1

Matrix 4

0	0	0
0	0	0
0	0	0

Rank = 0

Row echelon forms

1	1	1
0	1	0
0	0	1

Number of pivots = 3

1	1	1
0	0	1
0	0	0

Number of pivots = 2

1	1	1
0	0	0
0	0	0

Number of pivots = 1

0	0	0
0	0	0
0	0	0

Number of pivots = 0

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Solving System of Linear Equations

Reduced row echelon form

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Original matrix

5	1
4	-3

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Diagonal matrix

1	0
0	1

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $a = 3$
- $b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Diagonal matrix

1	0
0	1

Row echelon form

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $1a + 0b = 3$
- $0a + 1b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Row echelon form

Diagonal matrix

1	0
0	1

Reduced row echelon form

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $1a + 0b = 3$
- $0a + 1b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Row echelon form

Diagonal matrix

1	0
0	1

Reduced row echelon form

Systems of equations to matrices

Original system

- $5a + b = 17$
- $4a - 3b = 6$

Intermediate System

- $a + 0.2b = 3.4$
- $b = 2$

Solved system

- $1a + 0b = 3$
- $0a + 1b = 2$

Original matrix

5	1
4	-3

Upper diagonal matrix

1	0.2
0	1

Row echelon form

Diagonal matrix

1	0
0	1

Reduced row echelon form

Reduced row echelon form

Row echelon form

1	0.2
0	1

Reduced row echelon form

Row echelon form

1	0.2
0	1

→

0	1

Reduced row echelon form

Row echelon form

1	0.2
0	1

0	1

→

0	1

Reduced row echelon form

Row echelon form

1	0.2
0	1

0	1

0.2

0	1

Reduced row echelon form

Row echelon form

1	0.2
0	1

$\rightarrow$

0	1

0	1

x **0.2**

Reduced row echelon form

Row echelon form

1	0.2
0	1

→

0	1

0	1

x **0.2**

0	0.2

Reduced row echelon form

Row echelon form

1	0.2
0	1

0	1
1	0.2

0	1

x **0.2**

0	0.2

Reduced row echelon form

Row echelon form

1	0.2
0	1

$\rightarrow$

0	1

	0	1
x		**0.2**
	0	0.2

	1	0.2
-	0	0.2

Reduced row echelon form

Row echelon form

1	0.2
0	1

$\rightarrow$

0	1

	0	1
x		**0.2**
	0	0.2

	1	0.2
-	0	0.2
	1	0

Reduced row echelon form

Row echelon form

$$\begin{bmatrix} 1 & 0.2 \\ 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{array}{rcc} & 0 & 1 \\ \times & & \mathbf{0.2} \\ \hline & 0 & 0.2 \end{array} \qquad \begin{array}{rcc} & 1 & 0.2 \\ - & 0 & 0.2 \\ \hline & 1 & 0 \end{array}$$

Reduced row echelon form

Row echelon form

1	0.2
0	1

Reduced row echelon form

1	0
0	1

	0	1
x		**0.2**
	0	0.2

	1	0.2
-	0	0.2
	1	0

Reduced row echelon form

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Reduced row echelon form

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

- Is in row echelon form

Reduced row echelon form

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

- Is in row echelon form
- Each pivot is a 1

Reduced row echelon form

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

- Is in row echelon form
- Each pivot is a 1
- Any number above a pivot is 0

Reduced row echelon form

1	0	0	0	0
0	1	0	0	0
0	0	1	0	0
0	0	0	1	0
0	0	0	0	1

Rank 5

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Rank 3

- Is in row echelon form
- Each pivot is a 1
- Any number above a pivot is 0
- Rank of the matrix is the number of pivots

Reduced row echelon form

Reduced row echelon form

3	*	*	*	*
0	0	**2**	*	*
0	0	0	**-4**	*
0	0	0	**0**	0
0	0	0	0	**0**

Reduced row echelon form

Row echelon form

3	*	*	*	*
0	0	2	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

1	*	*	*	*
0	0	1	*	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Reduced row echelon form

Row echelon form

3	*	*	*	*
0	0	2	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

1	*	*	*	*
0	0	1	*	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Divide each row by the value of the pivot

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Reduced row echelon form

Row echelon form

3	*	*	*	*
0	0	2	*	*
0	0	0	-4	*
0	0	0	0	0
0	0	0	0	0

1	*	*	*	*
0	0	1	*	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Divide each row by the value of the pivot

Reduced row echelon form

1	*	0	0	*
0	0	1	0	*
0	0	0	1	*
0	0	0	0	0
0	0	0	0	0

Turn anything above a pivot to 0

Reduced row echelon form

Row echelon form

1	2	3
0	1	4
0	0	1

Reduced row echelon form

Row echelon form

1	2	3
0	1	4
0	0	1

Subtract 2 times the second row from the first one

Reduced row echelon form

Row echelon form

1	2	3
0	1	4
0	0	1

→

1	0	-5
0	1	4
0	0	1

Subtract 2 times the second row from the first one

Reduced row echelon form

Row echelon form

1	2	3
0	1	4
0	0	1

1	0	-5
0	1	4
0	0	1

Subtract 2 times the second row from the first one

Add 5 times the third row to the first one

Reduced row echelon form

Row echelon form

1	2	3
0	1	4
0	0	1

Subtract 2 times the second row from the first one

1	0	-5
0	1	4
0	0	1

Add 5 times the third row to the first one

1	0	0
0	1	4
0	0	1

Reduced row echelon form

Row echelon form

1	2	3
0	1	4
0	0	1

Subtract 2 times the second row from the first one

1	0	-5
0	1	4
0	0	1

Add 5 times the third row to the first one

1	0	0
0	1	4
0	0	1

Subtract 4 times the third row from the second one

Reduced row echelon form

Row echelon form

1	2	3
0	1	4
0	0	1

Subtract 2 times the second row from the first one

1	0	-5
0	1	4
0	0	1

Add 5 times the third row to the first one

1	0	0
0	1	4
0	0	1

Subtract 4 times the third row from the second one

1	0	0
0	1	0
0	0	1

Reduced row echelon form

Row echelon form

1	2	3
0	1	4
0	0	1

Subtract 2 times the second row from the first one

1	0	-5
0	1	4
0	0	1

Add 5 times the third row to the first one

1	0	0
0	1	4
0	0	1

Subtract 4 times the third row from the second one

Reduced row echelon form

1	0	0
0	1	0
0	0	1

DeepLearning.AI

Solving System of Linear Equations

Conclusion